

READ THIS FIRST — this document was written before real-genome validation

Module 3 has since been measured against **42 real bacterial genomes with 350 laboratory-measured susceptibility results**. On that test it scores **0.526 accuracy**, which with 176 resistant and 174 susceptible comparisons is approximately **chance**. No threshold changes this; the sweep peaks at 0.503.

Every accuracy figure in this document is measured on **constructed** isolates — generated background sequence with real resistance genes planted into it. Those figures are correct for constructed isolates and must not be presented as real-world performance.

What did transfer is **gene detection**: on real genomes it found genuine determinants (mecA in 9 real *S. aureus*, blaSHV in 5, plus blaNDM-1, blaVIM, blaCTX-M-15) at a 13.8% false-resistance rate, though it misses 73% of laboratory-confirmed resistance because 21 reference genes cannot cover real resistance. Full analysis: **reports/AUDIT_RESPONSE.md**.

TRIALSENSE • SEGUE 3.0 • INTERNAL BRIEFING

Module 3 Demonstration Guide

What to say, what to show, and what to answer. Written for presenters with an engineering background and no healthcare training. Every technical term is defined the first time it appears.

01 The problem we are solving

STATED PLAINLY

A pharmaceutical company that wants to sell a new antibiotic must first prove it works. That process takes roughly nine years and costs in the order of two hundred crore rupees.

The money is spent in stages. Laboratory work first, then testing in small groups of people, then testing in large groups of people. Costs rise sharply at each stage. The final stage is by far the most expensive.

Bacteria develop the ability to survive antibiotics. This is called **resistance**. It happens over years, and it spreads. When a bacterium becomes resistant to a drug, that drug stops working against it permanently.

WHERE THE MONEY IS LOST

A company can spend the full nine years developing a drug, only to find in the final testing stage that the bacteria it was designed for had already become resistant. The drug is sound. The target was wrong.

At that point the spending is unrecoverable. The information that would have prevented it — whether those bacteria were already resistant — existed from the start. Nobody checked it computationally, because there was no straightforward way to do so.

WHO THIS AFFECTS

WHO	THEIR PROBLEM	WHAT THEY NEED
R&D teams at pharmaceutical companies	They commit years of work before finding out whether the target was viable	A reliable answer on day one
The committee that approves budgets	They approve spending without knowing which candidates are already obsolete	A ranked comparison across the whole pipeline
Smaller Indian manufacturers	One failed programme is proportionally far more damaging than for a large multinational	Low-cost screening before commitment

OPENING LINE

“A company spends around nine years and two hundred crore developing an antibiotic. The most expensive way that fails is not safety — it is discovering late that the bacteria it was built for had already become resistant. Module 3 answers that question on day one, before the money is committed.”

02 What Module 3 does

THE INPUT

Module 3 takes the genetic sequence of one bacterial sample. A genetic sequence is a long text file containing only the letters A, C, G and T. It can be a sample we supply, or a file the user uploads.

THE PROCESSING

1. **Search for known resistance genes.** Certain short sections of bacterial DNA are known to make a bacterium resistant to particular antibiotics. We hold the published sequences for twenty-one of these. The module compares the uploaded sequence against each one.
2. **Score the whole sequence statistically.** Separately, a trained model reads the overall pattern of the DNA and estimates resistance across twelve antibiotic groups. This handles cases where no exact gene match is found.
3. **Combine the two.** Where the gene search finds a definite match, its result takes priority, because it is the more reliable of the two.

THE OUTPUT

Six separate reports, each answering a different question. They are not merged into a single score, because a single number would conceal cases where the six answers disagree.

REPORT	QUESTION IT ANSWERS
Runway	Will this drug still work at launch, not just today?
Rescue	Resistance was found — can it be worked around?
Portfolio	Which of our candidates performs best across all targets?
Unknown	Is there anything here we cannot identify?
Value	What is finding this early worth in rupees?
Spread	How quickly is this resistance moving through the population?

OVERVIEW LINE

“The input is a DNA file. The module searches it for twenty-one known resistance genes, scores it with a trained model, and produces six separate reports. We keep them separate on purpose, because a single combined number would hide the cases where the reports disagree — and those are the cases that matter.”

03 The six features in detail

1 • Runway STRONGEST FEATURE – LEAD WITH THIS

WHAT IT DOES

Projects how resistant the target bacteria will be in the year the drug would actually reach market, rather than how resistant they are today.

HOW IT WORKS

1. India's national health research body publishes resistance figures every year. We hold that data from 2016 to 2024.
2. The module fits a curve through those figures.
3. It extends the curve forward to the launch year the user enters.
4. It reports where the line crosses the level at which the drug would no longer be commercially worth selling.

WHY IT IS NEEDED

Development takes around a decade. Today's resistance figure is not the figure the finished drug will face. Every existing tool reports today's.

WHY IT MAKES US STAND OUT

Existing tools are built for hospital staff treating a patient now, so a present-day answer is correct for them. No tool we found projects forward to a specific product's launch year.

HOW IT REDUCES COST

It prevents a company spending nine years building a product for a market that will have disappeared by the time it is finished. That is the entire development budget, not a portion of it.

2 • Rescue MOST TECHNICALLY IMPRESSIVE

WHAT IT DOES

When resistance is found, it determines whether the programme can be saved by adding a second drug, or whether it must be abandoned.

HOW IT WORKS

1. The module identifies which gene caused the resistance.
2. It looks up how that gene works. Different genes block a drug by different physical means.
3. Some of those means can be counteracted by adding a second compound alongside the main drug. Others cannot.
4. It returns one of three results: fixable, partially fixable, or not fixable — with the specific reason.

WHY IT IS NEEDED

Two different genes can produce an identical result on screen — both reported as resistance to the same antibiotic group — while requiring opposite business decisions. One needs a few months of reformulation. The other needs the programme cancelled. Reporting only the group name loses that distinction entirely.

WHY IT MAKES US STAND OUT

Other tools report which antibiotic groups are affected. They do not report whether anything can be done about it. That second question is the one that determines what the company actually does next.

HOW IT REDUCES COST

It works in both directions. It prevents a team abandoning a programme that a small change would have saved, and it prevents a team spending months attempting a fix that cannot work.

3 · Portfolio

BEST VISUAL MOMENT

WHAT IT DOES

Tests every drug candidate the company holds against every bacterial target it is considering, and presents the results as a single grid.

HOW IT WORKS

1. The user selects several candidate drugs and several target bacteria.
2. The module runs the full analysis for every combination.
3. Each result is shown as a coloured cell — green for viable, red for blocked.
4. It ranks the candidates and recommends which to advance.

WHY IT IS NEEDED

A company does not hold one candidate. It holds several, and has funding to advance one or two. The decision is comparative, not individual.

WHY IT MAKES US STAND OUT

Every comparable tool analyses one bacterial sample at a time. This works at the level the funding decision is actually made.

HOW IT REDUCES COST

Producing this grid by laboratory testing takes weeks and costs lakhs. Here it takes seconds. The saving is direct and easy to verify.

4 • Unknown ANSWERS THE HARDEST QUESTION

WHAT IT DOES

Identifies sections of the DNA that appear to have come from another organism but do not match any gene we know about.

HOW IT WORKS

1. DNA acquired from a different organism has measurably different statistical properties from the organism's own DNA.
2. The module measures those properties in short overlapping sections along the sequence.
3. Sections already explained by an identified gene are set aside.
4. Anything remaining that still looks foreign is reported, along with its position.

WHY IT IS NEEDED

Every tool of this kind can only find what is already in its database. When such a tool reports nothing, that is indistinguishable from confirming that nothing is there. Those are very different statements.

WHY IT MAKES US STAND OUT

This is the only part of the system that reports the existence of something it cannot name. It is a deliberately weaker claim than a gene identification, and it is the honest one.

HOW IT REDUCES COST

It prevents a false clearance. A company proceeding on the assumption that nothing was found, when in fact nothing was recognised, is the most expensive possible error.

5 • Value HANDLE CAREFULLY – SEE PART 5

WHAT IT DOES

Converts the finding into a rupee figure: what it would have cost to discover the same problem later in the process.

HOW IT WORKS

1. The user enters the cost of each development stage and the probability of passing it.
Published figures are supplied as starting values.
2. The module identifies the stage at which this problem would normally have surfaced without a computational check.
3. It totals the spending committed by that point.
4. It weights that total by the probability the programme would have survived far enough to reach it.

WHY IT IS NEEDED

The people who approve budgets do not evaluate scientific findings. They evaluate financial cases. A finding that cannot be expressed in rupees does not reach them.

WHY IT MAKES US STAND OUT

Other tools produce a scientific result and stop. This translates it into the format required by the person who authorises the spending.

HOW IT REDUCES COST

Strictly speaking it does not reduce cost. It **measures** the reduction the other features produce, which is what makes the case for using them at all.

6 • Spread SHORT SEGMENT

WHAT IT DOES

Reports how quickly a resistance gene moves through a bacterial population.

HOW IT WORKS

1. Some resistance genes sit on a section of DNA that bacteria can pass directly to other bacteria, including bacteria of a different species.
2. Others are fixed in the main chromosome and pass only to the bacterium's own descendants.
3. The first kind spreads far faster. The module checks which kind each detected gene is.
4. It applies a corresponding adjustment to the Runway projection.

WHY IT IS NEEDED

The Runway projection is fitted to historical averages. A transferable gene spreads faster than that history implies, so the projection would otherwise be too optimistic.

WHY IT MAKES US STAND OUT

Resistance is normally reported as a fixed property. Reporting it as something with a rate of change is unusual, and it is what makes a forward projection meaningful.

HOW IT REDUCES COST

It corrects the timeline. A team planning against an unadjusted estimate is planning against a deadline that is further away than the real one.

04 Managing the volume of output

Module 3 produces six reports, each with several sections. Presenting all of it will exhaust the available time and the judges' attention. The following selection is deliberate.

SHOW THREE, HOLD THREE

FEATURE	IN THE DEMO	REASON
Rescue	Show	Establishes technical depth fastest
Portfolio	Show	Visual and immediately understood
Runway	Show	Backed by real government data
Spread	Mention in one line	Supports Runway, not independent
Unknown	Hold for questions	Requires setup to appreciate
Value	Hold for questions	Attracts scrutiny; better on request

RULE

One number per screen. Each feature has one figure worth stating aloud. Read that figure, explain what it means, move on. Do not narrate the rest of the screen — the judges can see it.

THE ONE NUMBER PER FEATURE

FEATURE	SAY ONLY THIS
Core scan	“Five resistance genes found. This one is the problem.”
Runway	“66% today, 84% by their launch year.”
Rescue	“Dead end — and here is the specific reason.”
Portfolio	“Sixteen combinations, five seconds.”
Unknown	“Clean, and the sample is typical of our training data.”
Value	“₹100 crore, but every input is adjustable.”
Spread	“Transferable, so the estimate is optimistic.”

AVOID

Do not read out the list of gene names. Do not explain the twelve antibiotic groups.
Do not walk through the settings panel. If a judge wants that detail they will ask, and answering on request is far stronger than volunteering it.

05 Anticipated questions

“You have no healthcare background. How do you know this is correct?”

“We did not decide any of the biology. Every gene sequence comes from the reference database maintained by the United States National Center for Biotechnology Information, with accession numbers you can look up. Every resistance relationship comes from two published databases. Our contribution is the engineering — the pipeline, the validation, and the decision logic built on top of established science.”

“How accurate is it?”

“0.84 on our most demanding test, where the model is evaluated on gene combinations it never saw during training. The easier test gives 0.90. We quote the harder number. In the deployed system the gene search overrides the model where it finds a definite match, and that removes every missed detection in our test set.”

“Is your data real?”

“The gene sequences are real and the national resistance figures are real. The DNA surrounding the genes in our sample files is constructed, because complete annotated genomes are far larger than a demonstration can carry. We state that in our documentation rather than leaving it to be discovered.”

“Tools for this already exist. Why build another?”

“Existing tools are built for hospital staff treating a patient today, and they are better at that than we are. They do not project to a launch year, they do not say whether resistance can be worked around, and they do not compare a portfolio of candidates. Those three questions determine whether a company funds a programme. That is the gap we built for.”

“Your cost saving figure — where does it come from?”

“It is a calculator, not a measurement. Every input is a slider seeded with a published default. We are not claiming to have saved anyone money. The claim we would defend is narrower: running this check costs effectively nothing, and not running it costs at minimum one failed development stage.”

“What happens with bacteria you have never seen?”

“Two things. The Unknown report flags DNA that looks acquired from elsewhere but matches nothing we hold — so we report its presence without claiming to identify it. And the system measures how similar the sample is to its training data, and says so, rather than sounding equally confident about everything.”

“What is the weakest part?”

“The statistical model. On its own it misses too much. We do not ship it on its own — the gene search overrides it. The underlying reason is our training data: the genes are real but the surrounding DNA is constructed. Replacing it with complete public genomes is the single change that would improve this most, and those genomes come with laboratory-confirmed results attached.”

“How would you improve it?”

“Complete real genomes from the public pathogen database. That fixes the training data and gives us laboratory-confirmed labels instead of ones we assign. Every other improvement is smaller than that one.”

IF YOU DO NOT KNOW AN ANSWER

Say “I do not know, and I would rather check than guess.” It costs nothing and protects the credibility of everything you have already said. A presenter who has admitted one limit is believed on the rest.

06 Presentation script

Approximately four minutes. Timings are cumulative.

0:00

SETUP

BEFORE YOU SPEAK

Case 3 already loaded. The ③ Resistance tab already open. Do not select anything while talking.

0:00

THE
PROBLEM

“A company spends around nine years and two hundred crore developing an antibiotic. The most expensive way that fails is not safety — it is discovering late that the bacteria it was built for had already become resistant. The information existed from the start. Nobody checked it.”

“Our user is an R&D team at an Indian pharmaceutical company. They are considering a drug against one of India's most common hospital infections. This is Module 3 analysing that bacteria's DNA.”

0:35

CORE SCAN

“It has read the DNA and found five resistance genes. This one is called blaNDM-1 — and it is worth noting that NDM stands for New Delhi. It was first identified in a patient here in 2008, and it is now across South Asian hospitals. It defeats the exact antibiotic group this company was planning to develop.”

1:05

RESCUE

CLICK

Rescue sub-tab

“Here is where we go past what other tools do. They tell you resistance exists. That is not a decision.”

“Bacteria block antibiotics by different physical means. Some of those can be counteracted by adding a second compound. If this were a different gene, that is exactly what we would recommend — a few months of reformulation, and the programme survives.”

“This gene works differently, and no available second compound counteracts it. Identical result on screen. Completely opposite business decision. One is a formulation change. This one means re-scoping a two hundred crore programme.”

2:00

PORTFOLIO

CLICK

Portfolio sub-tab — let the grid appear while you speak

“A company does not hold one candidate. It holds several and can fund one or two. Four candidates against four targets — sixteen combinations, five seconds. The same grid by laboratory testing is weeks of work.”

“And the answer is not the one they started with. The drug they were about to develop is blocked against three of the four. A different candidate is clear against all four.”

CLICK

Runway sub-tab

“Every other tool reports resistance today. A drug starting now reaches market around 2035, so today is the wrong year to ask about.”

“This is real data — India's national surveillance network, 2016 through 2024. Resistance has risen from 46% to 66%. Projected to 2035: 84%. It crosses the level where the drug stops being commercially worth selling around 2030 — five years before they would even launch.”

“And we tested the projection rather than just drawing it. We fitted it on data up to 2020 only, then asked it to predict the four years it had never seen. Mean error: 4.93 percentage points.”

“One last point, because it is the part I would want to hear.”

“We were training on constructed DNA. We replaced it with real published sequences from the NCBI reference database — and our headline accuracy fell from 0.99 to 0.90. We kept the change. That gap was the constructed data flattering us, and we would rather report the number that is true.”

“The product got better, not worse, because gene detection on real sequences is exact — so we let it override the statistical model. Missed detections went from five to zero. Our honest figure on the hardest test is 0.84.”

THEN

Stop talking. Let them ask.

07 Figures to know without looking

HONEST ACCURACY 0.763 hardest held-out test	FORECAST ERROR 4.93 pp 184 unseen predictions	MISSED DETECTIONS 0 in the shipped system	GENE SEQUENCES 21 real, from NCBI
GENE SEARCH 12 / 12 samples, no false matches	REAL DATA SPAN 2016–24 ICMR national figures	VALIDATION CHECKS 8 each a different question	AUTOMATED TESTS 51 all passing

PHRASES TO AVOID, AND THEIR REPLACEMENTS

DO NOT SAY	SAY INSTEAD
“We use real bacterial genomes”	“Real gene sequences, constructed surrounding DNA”
“Laboratory validated”	“We assign the sample contents, so the labels are ours”
“It discovers new resistance genes”	“It flags DNA it cannot identify”
“We saved them ₹45 crore”	“The calculator estimates ₹45 crore”
“99% accurate”	“0.84 on the hardest test”

FINAL INSTRUCTION

If time runs short, present Rescue, then Portfolio, then the closing paragraph about accuracy falling. Those three carry the module on their own.

TrialSense is a research screening prototype for pharmaceutical R&D. It is not a diagnostic tool and not a clinical decision-support system. No output should be used to treat anyone.